

Optimal Control of Solar Sail under Measurement Uncertainties

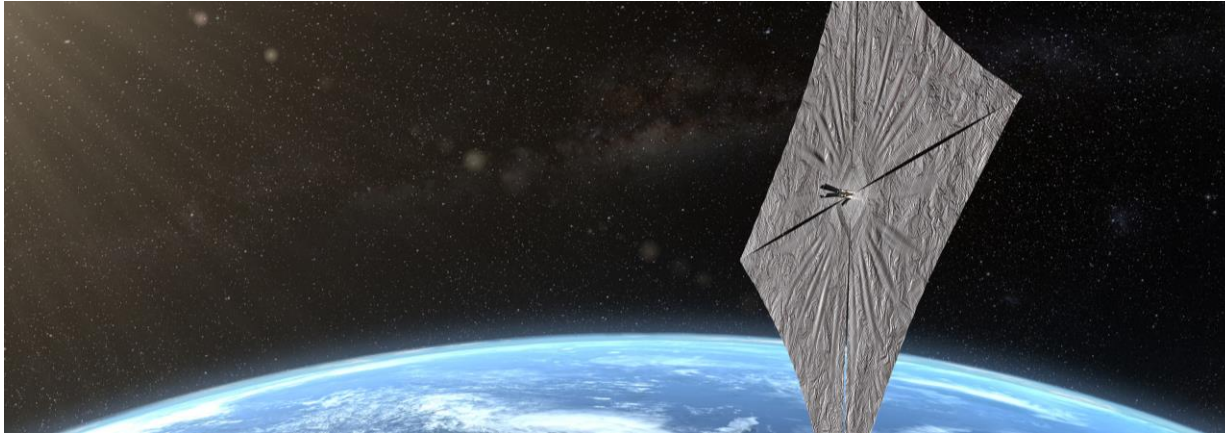
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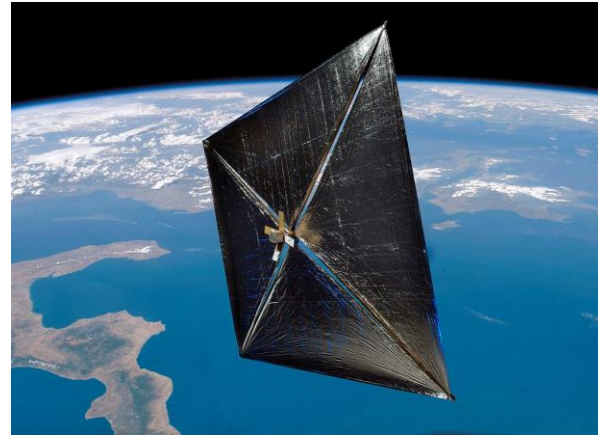
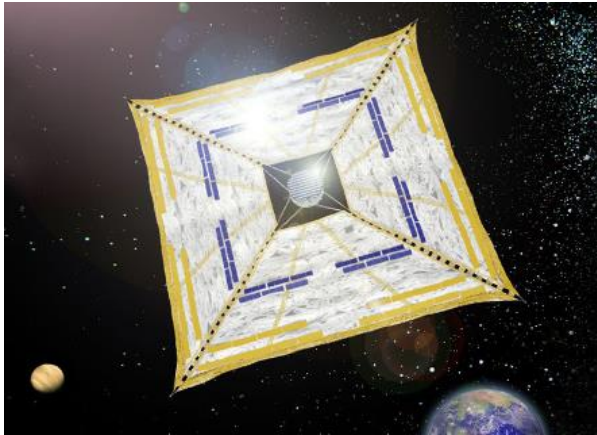
Motivation – Why are Solar Sails Important?



Solar Sails provide propellant-free propulsion which can enable long duration missions.

There is pre-existing literature on guidance and control for solar sails:

- Tsuda et al. demonstrated successful use of solar sails beyond Earth's orbit with the IKAROS mission (bottom left).
- Johnson et al. and Spencer et al. showcased the effectiveness of solar sailing for CubeSats and small spacecraft via the NanoSail-D and LightSail missions (top and bottom right).
- This project leverages and adapts the robust control framework established by Oguri et al., simplifying it for specific mission requirements.



Problem Formulation

Objective :

$$\min_u \int_0^{t_f} 1 dt$$

s.t.

$$\dot{x} = f(x) + g(x, u)$$

$$u = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$0 \leq \alpha \leq \alpha_{max}$$

Where

$$f(x) = \begin{bmatrix} v \\ \frac{-\mu r}{d^3} \end{bmatrix}, d = \|r\|_2$$

$$g(x, u) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \hat{r} & \hat{h} \times \hat{r} & \hat{h} \end{bmatrix} \begin{bmatrix} a_1(u) \\ a_2(u) \\ a_3(u) \end{bmatrix} \frac{\sigma G_0 \cos \alpha}{d^2}$$

$$a_1(u) = C_1 \cos^2 \alpha + C_2 \cos \alpha + C_3$$

$$a_2(u) = -(C_1 \cos \alpha + C_2) \sin \alpha \sin \beta$$

$$a_3(u) = -(C_1 \cos \alpha + C_2) \sin \alpha \cos \beta$$

Optimal Control

Optimality Conditions

$$H = 1 + \lambda^T (f(x) + g(x, u))$$

$$\dot{\lambda} = \frac{-\partial H}{\partial x} = -\lambda^T \left(\frac{\partial f}{\partial x} + \frac{\partial g}{\partial x} \right)$$

Boundary Conditions

$$\delta t_0 = 0, \delta x_0 = 0$$

$$\psi_f = x(t_f) - x_f = 0$$

$$\lambda^T \dot{x}_{t_f} + H_{t_f} = 0$$

Optimal Control

$$p = - \begin{bmatrix} 0 \\ I_{3 \times 3} \end{bmatrix}^T \lambda = - \begin{bmatrix} \lambda_4 \\ \lambda_5 \\ \lambda_6 \end{bmatrix}$$

$$p_x = -\lambda_4, p_y = -\lambda_5, p_z = -\lambda_6$$

$$u^* = \begin{bmatrix} \text{Median}(0, \alpha^*, \alpha_{max}) \\ \arctan2(-p_y, p_z) \end{bmatrix}$$

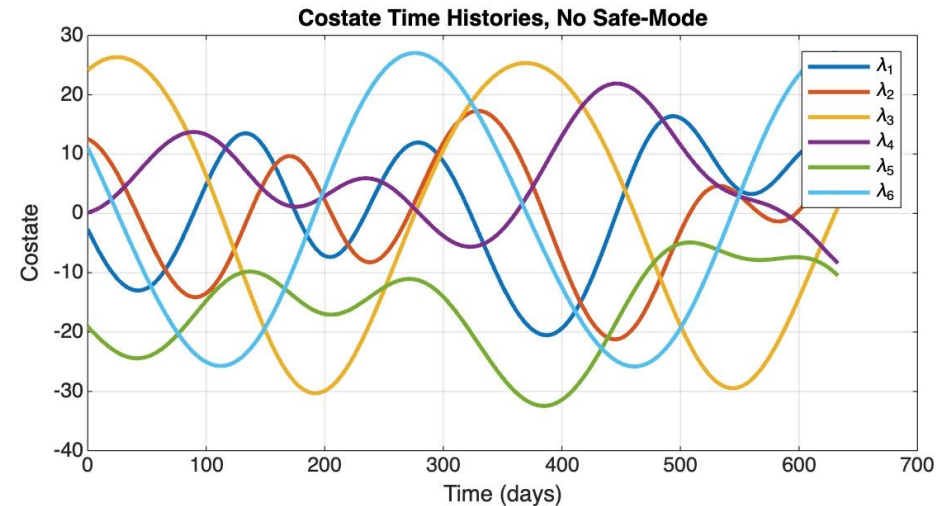
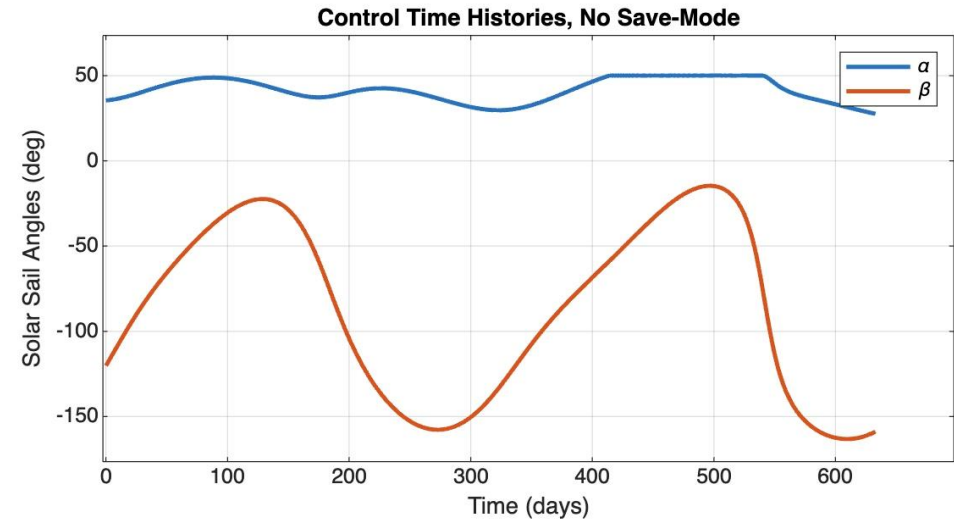
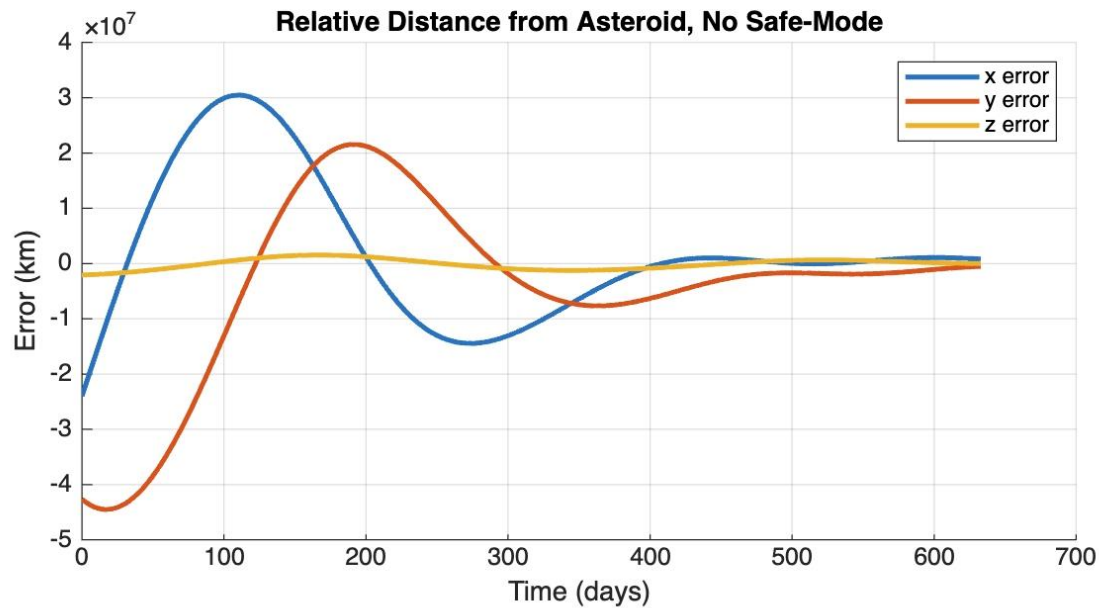
$$\alpha^* = \arctan\left[\frac{1}{4}(-3\tau + \sqrt{8 + 9\tau^2})\right]$$

$$\tau = \frac{p_x}{\sqrt{p_y^2 + p_z^2}}$$

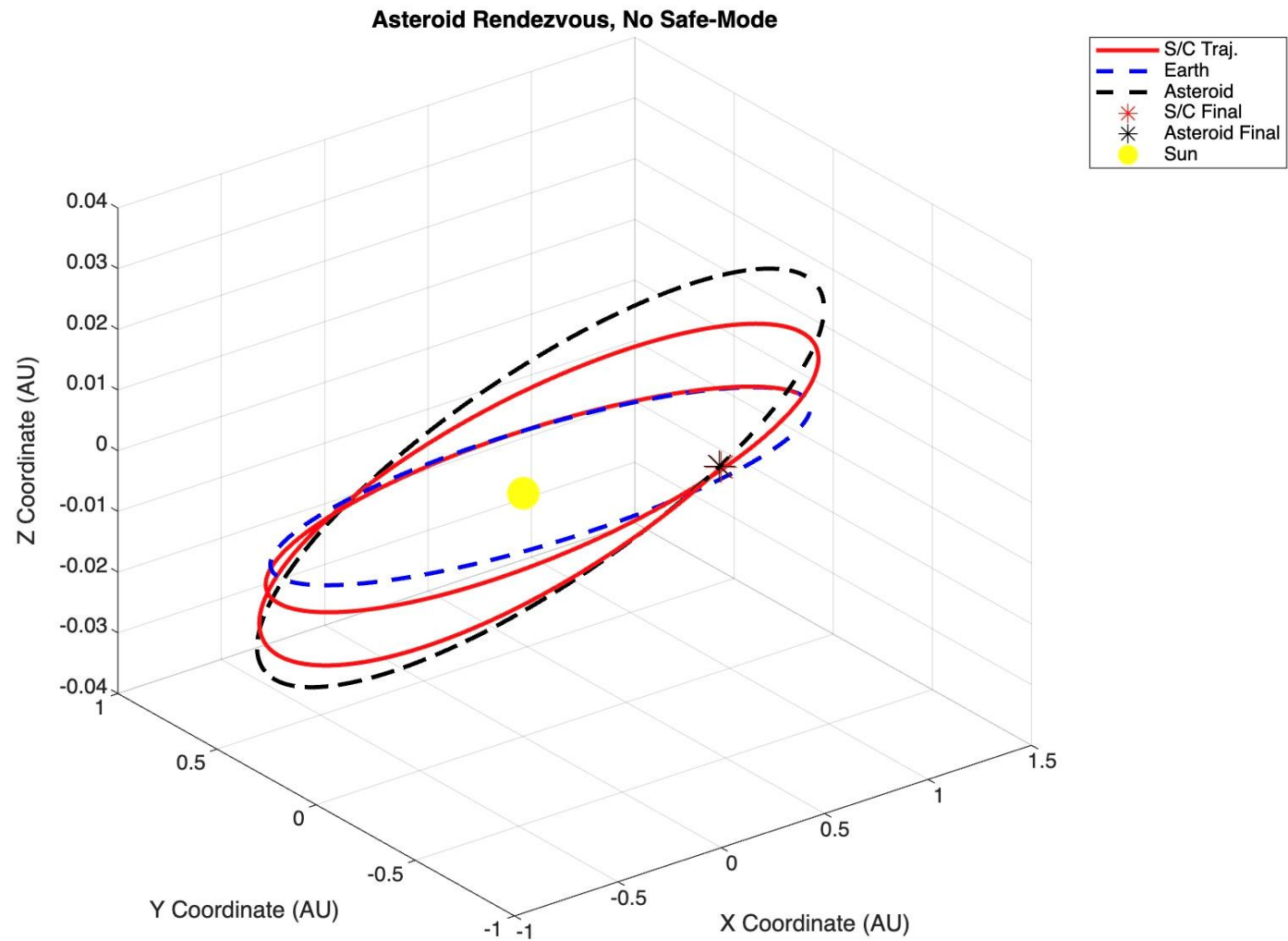
Assuming perfect reflectivity of solar sail.
Under imperfect reflectivity, Taylor Series Expansion must be used.*

Perfect reflectivity will make $C1 = 2, C2 = 0, C3 = 0$.

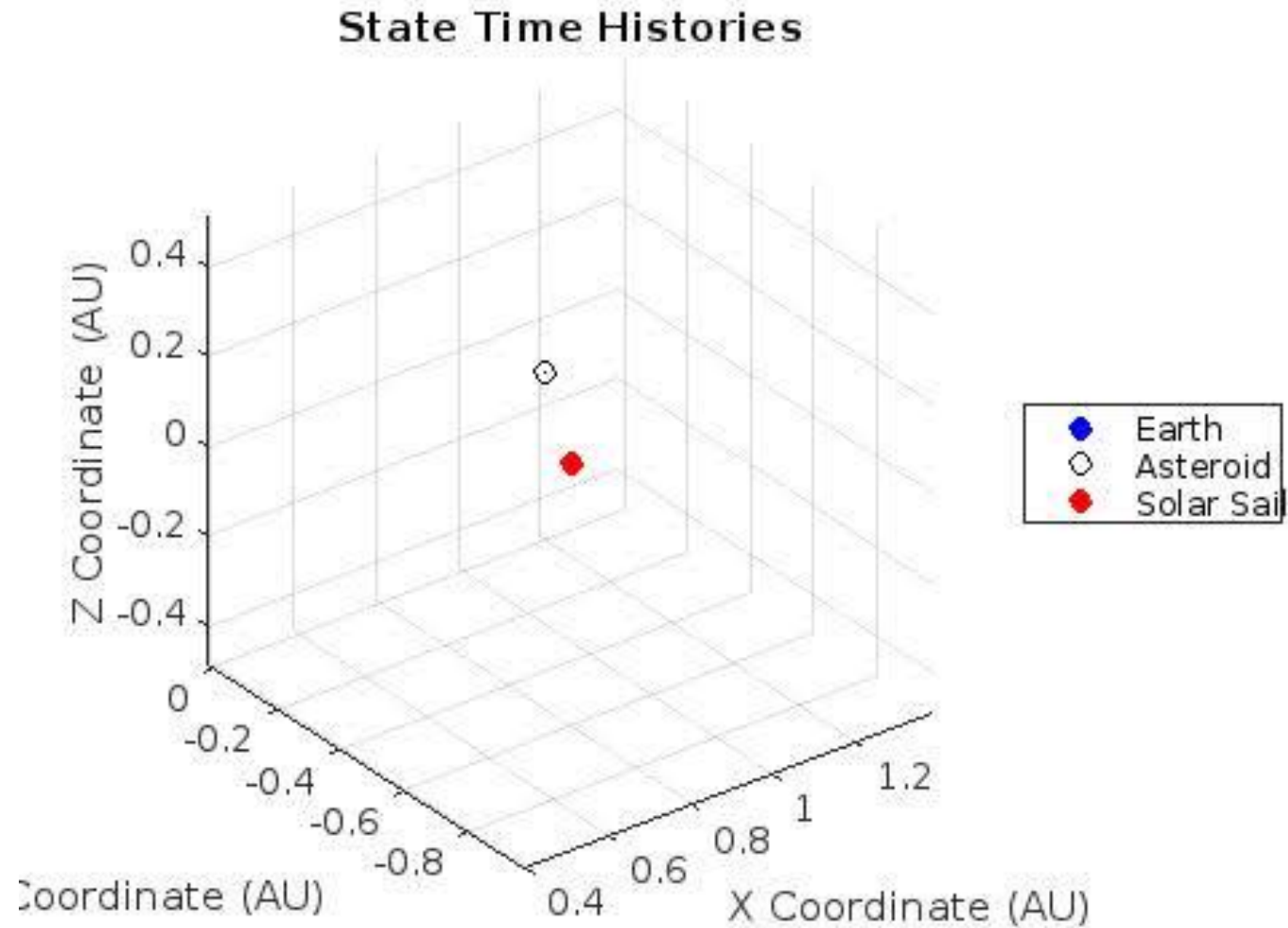
No Safe-Mode Event, No Measurement Noise



No Safe-Mode Event, No Measurement Noise



No Safe-Mode Event, No Measurement Noise



State Estimation

$$\hat{x}_k^- = A\hat{x}_{k-1} + Bu_k$$

$$P_k^- = AP_{k-1}A^T + Q$$

$$K_k = \frac{P_k^- C^T}{CP_k^- C^T + R}$$

$$\hat{x}_k = \hat{x}_k^- + K_k(y_k - c\hat{x}_k^-)$$

$$P_k = (I - K_k C)P_k^-$$

$$\hat{x}_0 = x_0$$

$$P_0 = \text{diag}([\sigma_{m_x}^2, \sigma_{m_y}^2, \sigma_{m_z}^2, \sigma_{m_{vx}}^2, \sigma_{m_{vy}}^2, \sigma_{m_{vz}}^2])_{6 \times 6}$$

$$A_{6 \times 6} = \frac{\partial \bar{f}}{\partial x} = \frac{\partial \bar{f}_0}{\partial x} + \frac{\partial \bar{g}}{\partial x}$$

$$Q = \text{diag}([\sigma_{p_x}^2, \sigma_{p_y}^2, \sigma_{p_z}^2, \sigma_{p_{vx}}^2, \sigma_{p_{vy}}^2, \sigma_{p_{vz}}^2])_{6 \times 6}$$

$$y = Cx + Du$$

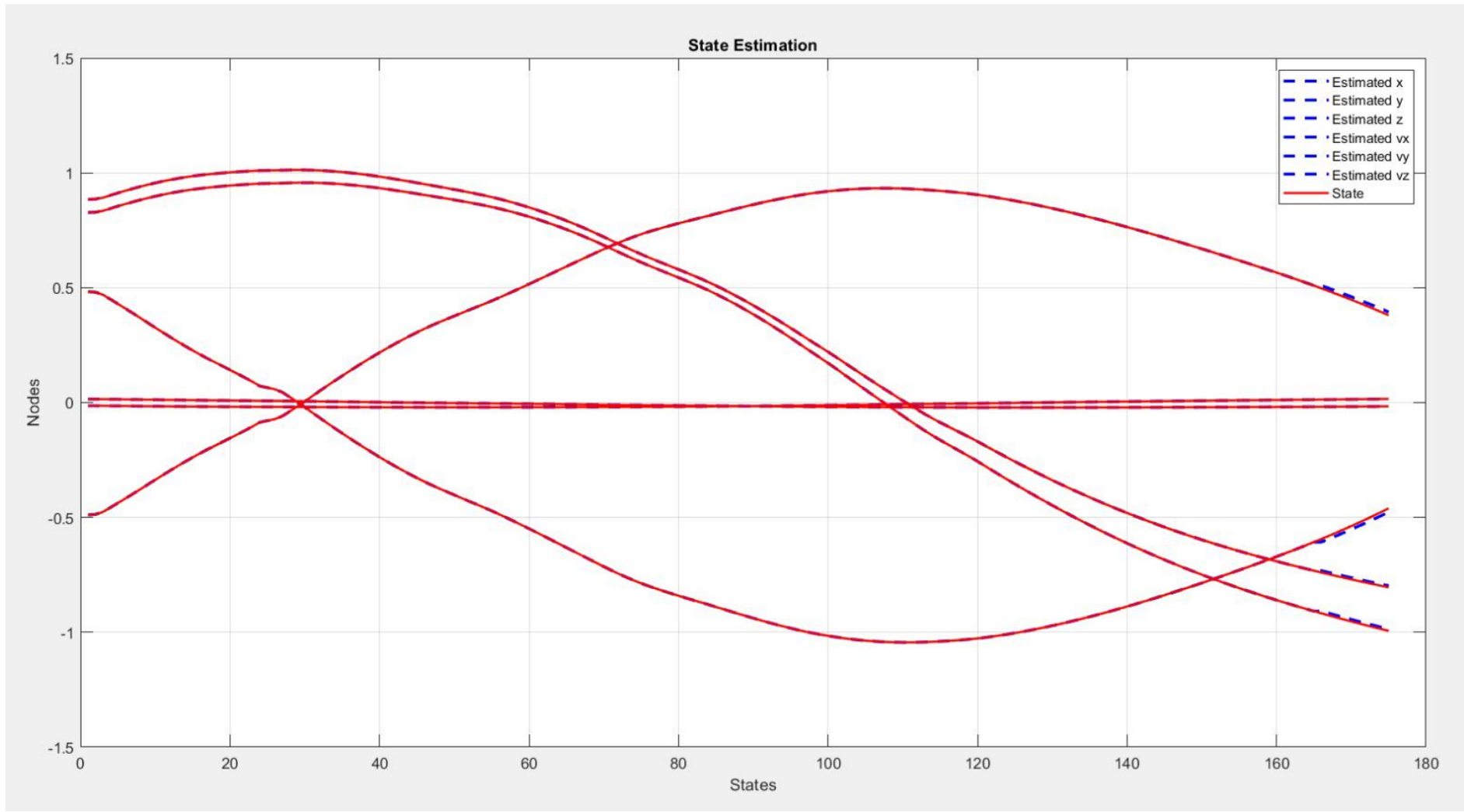
$$C = I_6, D = 0$$

$$R = \text{diag}([\sigma_{m_x}^2, \sigma_{m_y}^2, \sigma_{m_z}^2, \sigma_{m_{vx}}^2, \sigma_{m_{vy}}^2, \sigma_{m_{vz}}^2])_{6 \times 6}$$

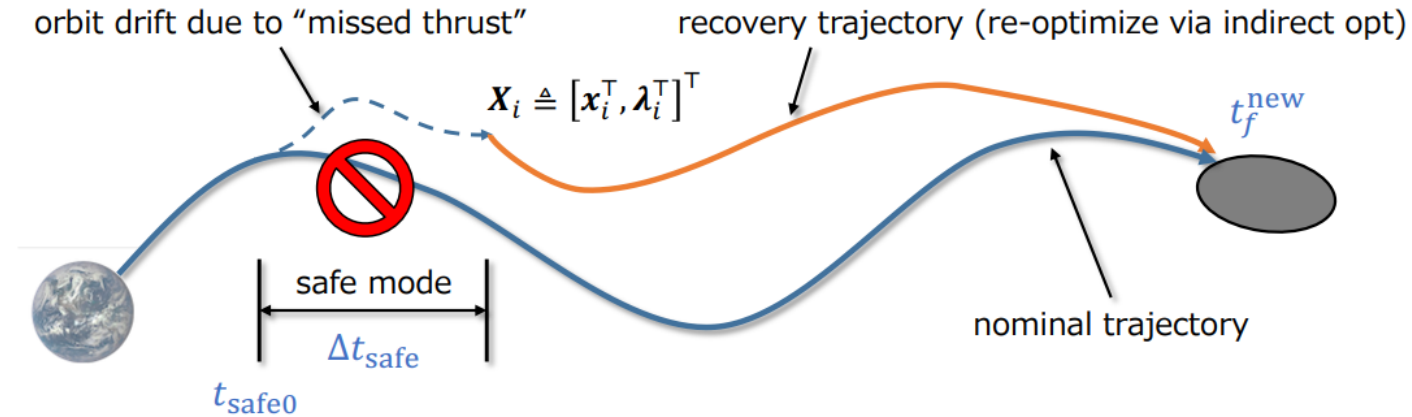
- Estimation at discrete time steps.
- Measurements created by adding noise to the dynamics.
- Assume we get measurements for both position and velocity.



State Estimation



Safe-Mode Event

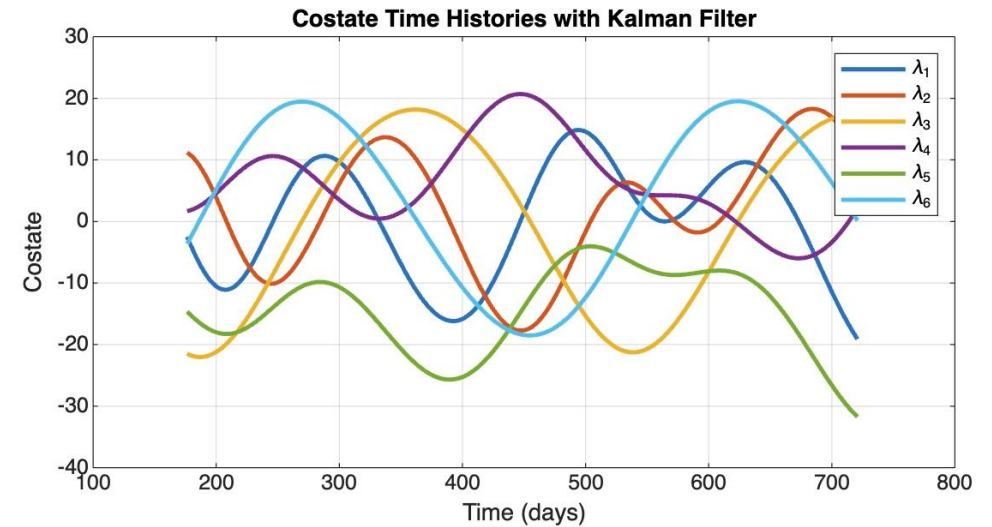
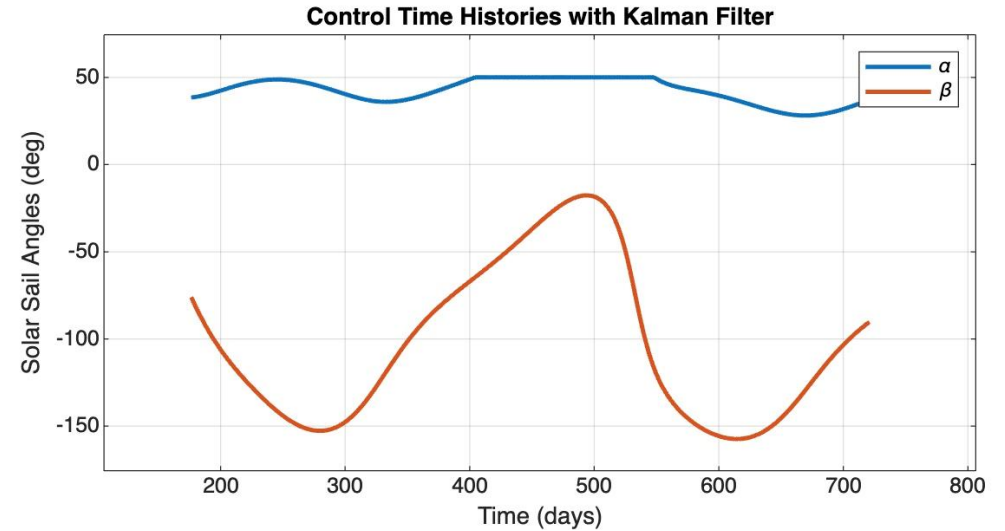
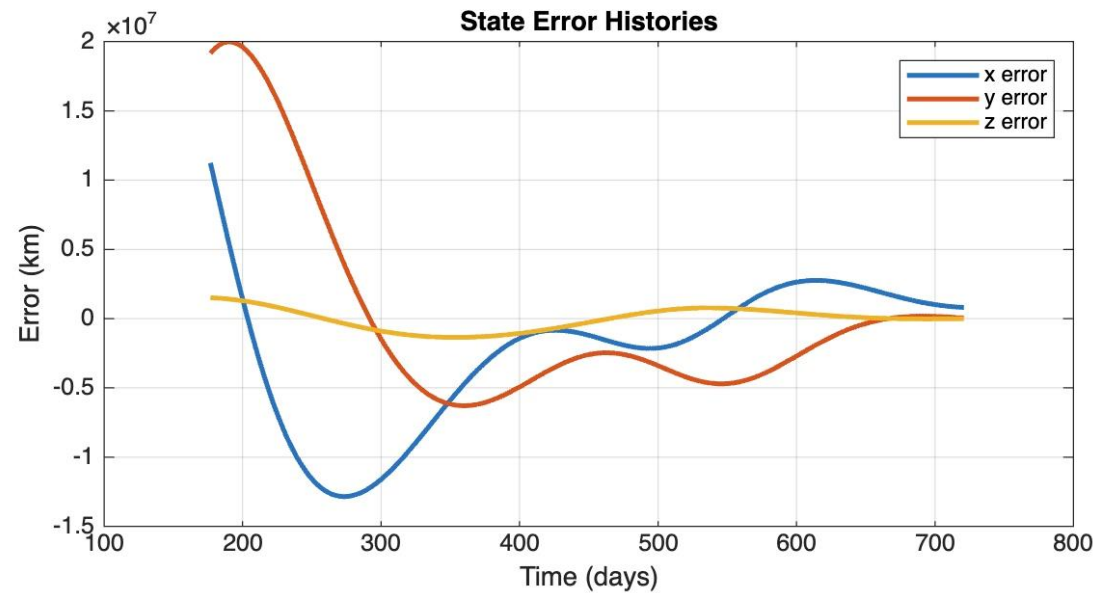


Safe-mode also called missed-thrust event: spacecraft experiences unexpected hardware issues, so it cannot perform originally planned optimal control.

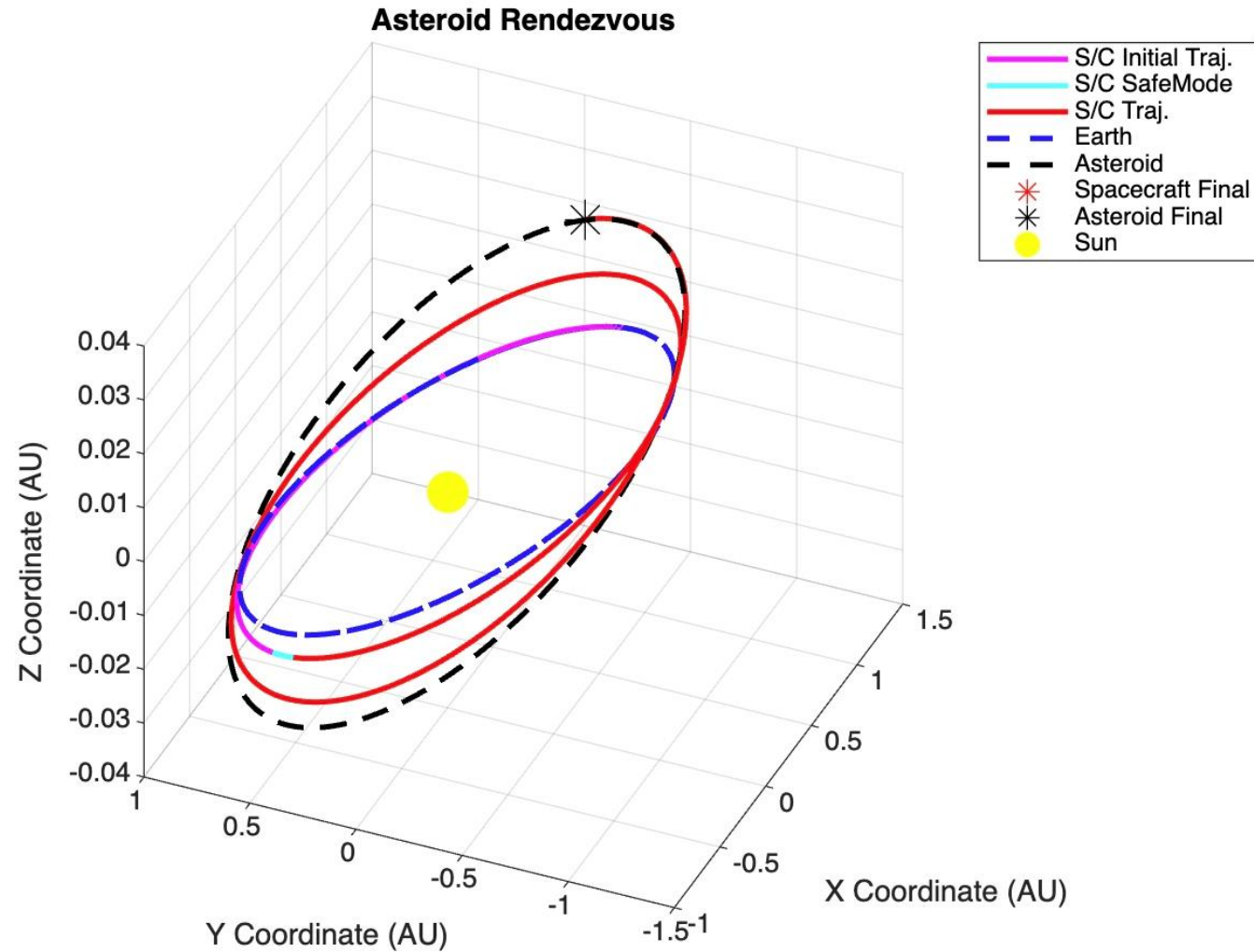
- Safe mode duration: 8 days
- Coasts with fixed control angle ("thrust" cannot be shut off).
- Time to recalculate trajectory should be reasonable for onboard spacecraft capabilities.

Figure taken from: Oguri, K. "Solar Sailing Trajectory Indirect Optimization: Application to NEA Scout"

Safe-Mode Event with Measurement Noise



Safe-Mode Event with Measurement Noise



Thank You